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Department of Electronic Systems

Examination paper forTFE4141 Design of Digital Systems 1

Examination date: December 20, 2022

Examination time (from-to): 15:00 – 19:00

Permitted examination support material: A / All printed and hand-written support material is allowed. All calculators are allowed.

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OTHER INFORMATION

Get an overview of the question set before you start answering the questions.

Read the questions carefully and make your own assumptions. If a question is unclear/vague, make your own assumptions and specify them in your answer. Some design tools (such as Vivado) may behave differently than what is specified in the VHDL standard. Answers to the questions in this exam should reflect the standard as it is presented in lectures and textbook. Only contact academic contact in case of errors or insufficiencies in the question set. Address an invigilator if you wish to contact the academic contact. Write down the question in advance.

InsperaScan: For questions **13** and **14** it is possible to submit the entire/some parts of the answer as handwritten sheets. At the bottom of the question you will find a seven-digit code. Fill in this code in the top left corner of the sheets you wish to submit. We recommend that you do this during the exam. If you require access to the codes after the examination time ends, click “Show submission”.

Weighting: The total number of achievable points is 100. The maximum number for each question is given on each question cover page.

Notifications: If there is a need to send a message to the candidates during the exam (e.g. if there is an error in the question set), this will be done by sending a notification in Inspera. A dialogue box will appear. You can re-read the notification by clicking the bell icon in the top right-hand corner of the screen.

Withdrawing from the exam: If you become ill or wish to submit a blank test/withdraw from the exam for another reason, go to the menu in the top right-hand corner and click “Submit blank”. This cannot be undone, even if the test is still open.

Access to your answers: After the exam, you can find your answers in the archive in Inspera. Be aware that it may take a working day until any hand-written material is available in the archive.

1 Logic signals (Correct answer 2 points, wrong answer -1 point, no answer 0 points)

Given the code below, first some signal declarations, and then two logic statements further down in the code. Assume that A and B have been high for a long time. Then a 1 ns low pulse appears on A before it returns to high. Which of the alternatives below is correct with respect to the resulting values on signals C and D?

```
signal A, B, C, D : STD_LOGIC := '1';
...
C <= A and B after 2 ns;
D <= transport A xnor B after 2 ns;
```

Select one alternative:

- Both of the other alternatives are wrong.
- D stays high while C has a 1 ns low pulse.
- C stays high while D has a 1 ns low pulse.

Maximum marks: 2

2 Bit pattern (Correct answer 2 points, wrong answer -1 point, no answer 0 points)

What is the resulting bit pattern on DataOut of the following VHDL statement?

```
DataOut <= "10011100" sra 3;
```

Select one alternative:

- "11110011"
- "00010011"
- "11100000"

Maximum marks: 2

3 Type of circuit

Given the code below. What type of circuit does it model.

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;

entity QuestionCircuit is
    port (  clk : in std_logic;
            sdi : in std_logic;
            sdo : out std_logic);
end QuestionCircuit;

architecture rtl of QuestionCircuit is
    signal data_n: std_logic_vector(3 downto 0);
    signal data_r : std_logic_vector(3 downto 0);
begin

    process(clk)
    begin
        if(clk'event and clk='1') then
            data_r <= data_n;
        end if;
    end process;

    process(sdi, data_r)
    begin
        data_n <= data_r(2 downto 0) & sdi;
    end process;

sdo <= data_r(3);

end rtl;
```

Select one alternative:

- A shift register
- A regular register
- A scan chain for use in production testing

Maximum marks: 2

4 Subtype (Correct answer 2 points, wrong answer -1 point, no answer 0 points)

Given the definition of the type std_ulogic:

```
TYPE std_ulogic IS ( 'U', -- Uninitialized
                     'X', -- Forcing Unknown
                     ‘0’, -- Forcing 0
                     ‘1’, -- Forcing 1
                     ‘Z’, -- High Impedance
                     ‘W’, -- Weak Unknown
                     ‘L’, -- Weak 0
                     ‘H’, -- Weak 1
                     ‘-‘, -- Don't care );
```

Assume that you have defined the following subtype
SUBTYPE my_logic IS std_ulogic range 'X' to 'Z';

You then try to assign a sequence of the following values to a signal of type my_logic: '0', 'Z' 'L'. How many of these are you allowed to assign to the signal?

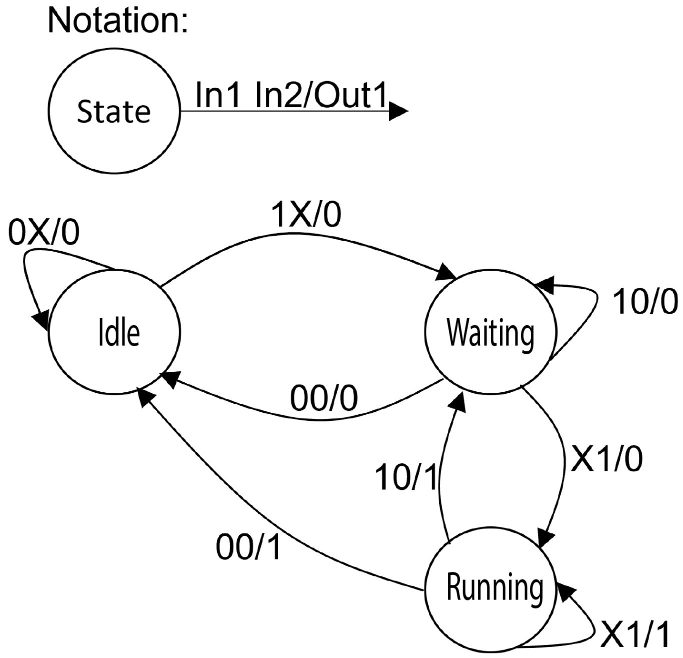
Select one alternative:

- All three
- Only one
- Two

Maximum marks: 2

5 State Machine (Correct answer 2 points, wrong answer -1 point, no answer 0 points)

Given the state diagram below and the three alternative sequences of input values on In1 and In2. Assume that In1 and In2 change value on the negative flank of a clock signal, while the state machine changes state on the positive flank of the same clock signal. Assume further that the state machine starts in state Idle. Which sequence of values will result in a single one clock period long positive pulse on Out1 (which otherwise is low)?



Sequence A		Sequence B		Sequence C	
In1	In2	In1	In2	In1	In2
0	0	1	0	0	1
0	1	1	0	1	1
1	1	1	1	0	1
1	0	1	1	1	0
0	0	0	0	1	0

Select one alternative:

- Sequence C
- Sequence B
- Sequence A

Maximum marks: 2

6 States in state machine (Correct answer 2 points, wrong answer -1 point, no answer 0 points)

Given the FSM in the previous task. What is the minimum number of flip-flops in the synthesized circuit?

Select one alternative:

- 1
- 2
- 3

Maximum marks: 2

7 State machine coding (Correct answer 2 points, wrong answer -1 point, no answer 0 points)

Given the state machine in the two previous tasks and the VHDL code below. The code shows a part of the combinatorial process in the recommended format but with modified state names. Which state in the state diagram does QuestionState correspond to?

```
when QuestionState =>
    Out1 <= '0';
    if In1 = '0' and In2 = '0' then
        next_state <= OtherState1;
    elsif In1 = '1' and In2 = '0' then
        next_state <= QuestionState;
    else
        next_state <= OtherState2;
    end if;
```

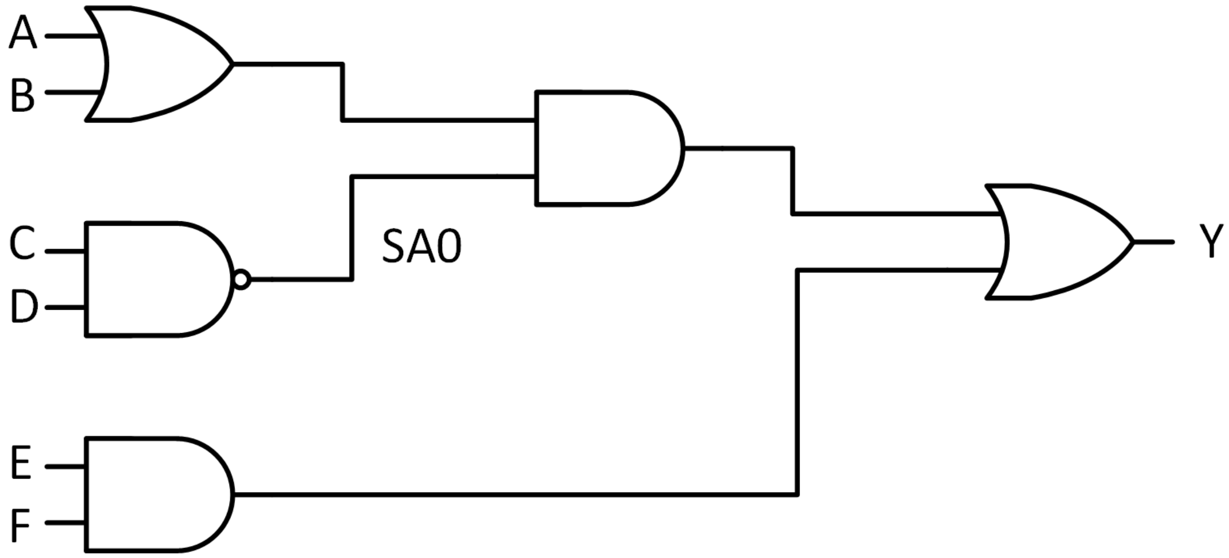
Select one alternative:

- Waiting
- Idle
- Running

Maximum marks: 2

8 Test patterns (Correct answer 2 points, wrong answer -1 point, no answer 0 points)

Given the circuit below. What pattern on the primary inputs can NOT be used to detect the output of the NAND gate stuck at 0?



Select one alternative:

- ABCDEF = 110111
- ABCDEF = 100110
- ABCDEF = 111001

Maximum marks: 2

9 True or false (Correct answer 2 points, wrong answer -1 point, no answer 0 points)

Which of the following statements is FALSE?
Select one alternative:

- A signal of type std_logic can not have more than one driving sources.
- A delay-fault found during production testing may be caused by variations in doping in a circuit.
- In many digital systems, accessing the memory subsystem consumes a major part of the total power.

Maximum marks: 2

10 True and false (Correct answer 2 points, wrong answer -1 point, no answer 0 points)

Which one of the following statements is true:
Select one alternative:

- Generic constants are included in the port list of an entity.
- Generic constants are mainly used for debugging of code.
- Generic constants can simplify reuse of entities between designs.

Maximum marks: 2

11 Inferred Latches (Maximum 12 points)

The code below will infer latches during synthesis. Answer the following questions in the text box below.

- Why is inferred latches in general a problem?
- Identify the parts of the code that results in latches and explain what the problem with the code is.
- Rewrite the part of the code that results in latches in such a way that the latches are avoided (you do not have to repeat the rest of the code).

```
library ieee;
use ieee.std_logic_1164.all;
use ieee.numeric_std.all;

entity InferredLatches is Port ( a, b, c, d : in std_logic_vector(15 downto 0);
                                 operation : in std_logic_vector( 1 downto 0);
                                 z : out std_logic_vector(15 downto 0);
                                 y : out std_logic_vector(15 downto 0) );
end InferredLatches;

architecture rtl of InferredLatches is
    constant A_ADD_B : std_logic_vector(1 downto 0):= "00";
    constant A_SUB_B : std_logic_vector(1 downto 0):= "01";
    constant C_ADD_D : std_logic_vector(1 downto 0):= "10";
    constant C_SUB_D : std_logic_vector(1 downto 0):= "11";
    signal result1 : signed(15 downto 0);
    signal result2 : signed(15 downto 0);

begin

    process(a,b,c,d, operation)
    begin
        case(operation) is
            when A_ADD_B =>
                result1 <= signed(a) + signed(b);
            when A_SUB_B =>
                result2 <= signed(a) - signed(b);
            when C_ADD_D =>
                result1 <= signed(c) + signed(d);
            when others => -- C_SUB_D
                result2 <= signed(c) - signed(d);
         end case;
     end process;

     y <= std_logic_vector(result1);
     z <= std_logic_vector(result2);

 end rtl;
```

Fill in your answer here

Format B I U x_2 x^2 I_x Ω Σ

Words: 0

Maximum marks: 12

12 Voltage and frequency scaling (Maximum 10 points)

Explain why frequency scaling can save power but not energy, while voltage and frequency scaling can save both power and energy.

Fill in your answer here

Format

Σ

Words: 0

Maximum marks: 10

13 RSA cryptography (Maximum 14 points)

Based on your experience from the term project, and general understanding of the RSA algorithm, what technique(s) would you use to speed up encryption of large messages?

You do not have to go into implementation details but should include a simplified block diagram / micro architecture to assist your explanation. This can be drawn on a separate sheet of paper to be marked with the control code and handed in at the end of the exam.

Fill in your answer here

Format

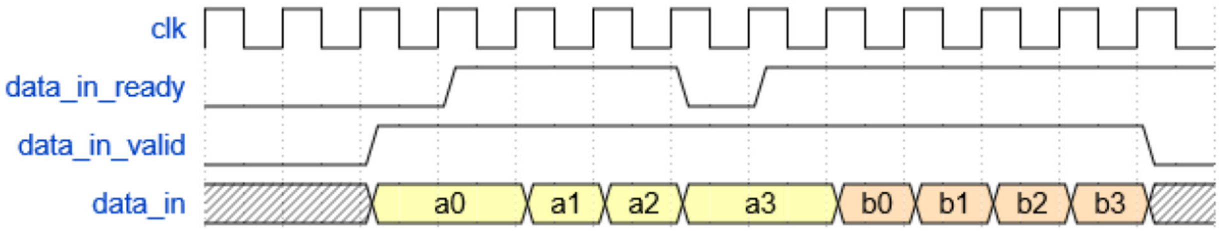
Σ

Words: 0

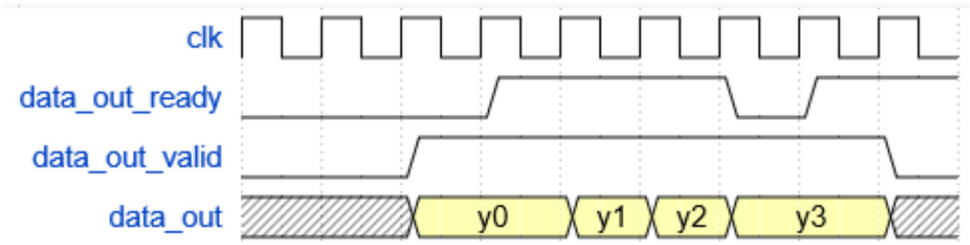
Maximum marks: 14

14 Design task (maximum 44 points)

You shall design a circuit that adds eight 16-bit signed numbers that arrive on a 64-bit wide data bus, with four numbers arriving in parallel each bus cycle. The final result shall be written to a separate result bus. A valid-ready protocol similar to the one used in the term project controls the communication on the two buses. See the following figure.



Input interface.



Output interface.

Solve the tasks below, using the the answer box as much as possible, but also by drawing and writing on separate sheets of paper when necessary. These should be marked with the control code and handed in at the end of the exam.

a) (max 10 points)
Draw a micro-architecture of the adder circuit for a design that calculates the result as quickly as possible. You can use as many adders and other required circuits as you want, but fewer is better as long as you do not increase the total time required to generate the final result. Indicate specifically what bit-width you need to have on the signals connecting modules in your design.

b) (max 8 points)
Consider a situation where the entity receiving the addition results is only ready to receive new data every five clock cycles. The entity giving you input data is able to give you four new numbers every clock cycle. Draw a timing diagram that shows how two complete transfers of two times eight input data and two results can be executed.

c) (max 14 points)
Write VHDL-code for the data path described in a). In case you need control signals, you can for now simply assume that they are generated outside your current design. You should add comments to your code explaining what they do, however.
Describe all additional assumptions you make while writing the VHDL-code. In case you do not manage to complete all parts you should explain what is missing and describe it as "future work". This will not give full score, but still add to your points.

d) (max 12 points)
Write VHDL-code for a circuit that allows your adder to follow the valid-ready protocol.

Describe all additional assumptions you make while writing the VHDL-code. In case you do not manage to complete all parts you should explain what is missing and describe it as "future work". This will not give full score, but still add to your points.

Fill in your answer here

Format

Words: 0

Maximum marks: 44